

III.4 Klasická limita a kvantové korekce

→ pro nerelativistickou bezstrukturální částici:

$$\Gamma(E) = V \cdot A \cdot E^{3/2} \quad A = \frac{1}{h^3} \frac{4}{3} \pi (2m)^{3/2}$$

fugacita $z := e^{\beta\mu} \leftarrow$ očekávání klasické pro $z \ll 1$

protože $\langle n_i \rangle = \frac{1}{e^{\beta\epsilon_i} \mp 1} \approx e^{-\beta(\epsilon_i - \mu)} = z e^{-\beta\epsilon_i} \ll 1$ chceme

$\Rightarrow$ ideální plyn

$$P = A \int_0^\infty dE E^{3/2} \frac{1}{e^{\beta(E-\mu)} \mp 1} = A \beta^{-5/2} \int_0^\infty dx \frac{x^{3/2}}{z^{-1} e^x \mp 1}$$

$$P = \frac{3}{2} A \int_0^\infty dE E^{1/2} \frac{1}{e^{\beta(E-\mu)} \mp 1} = \frac{3}{2} A \beta^{-3/2} \int_0^\infty dx \frac{x^{1/2}}{z^{-1} e^x \mp 1}$$

→ vyjádříme pomocí funkce:

$$g_\nu^\pm(z) := \frac{1}{\Gamma(\nu)} \int_0^\infty dx \frac{x^{\nu-1}}{z^{-1} e^x \mp 1}$$

• rozvoj pro $z \approx 0$ dostaneme:

$$g_\nu^\pm(z) \approx \sum_{n=1}^\infty (\pm 1)^{n-1} \frac{z^n}{n^\nu} = \begin{cases} z + \frac{z^2}{2^\nu} + \dots & \text{bosony} \\ z - \frac{z^2}{2^\nu} + \dots & \text{fermiony} \end{cases}$$

• pro $z=1$

$$g_\nu^\pm(1) = \sum_{n=1}^\infty \frac{1}{n^\nu} = \zeta(\nu) \quad \text{Riemannova zeta funkce}$$

$\Rightarrow$ přepíšeme dostaneme:

$$\frac{P}{k_B T} = A \Gamma\left(\frac{5}{2}\right) (k_B T)^{3/2} g_{5/2}^\pm(z)$$

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$$A \Gamma\left(\frac{5}{2}\right) (k_B T)^{3/2} = \frac{1}{h^3} \frac{4}{3} \pi (2m)^{3/2} \frac{5}{2} \cdot \frac{3}{2} \cdot \underbrace{\sqrt{\pi}}_{\Gamma(1/2)} = \Lambda^{-3}$$

$$\Lambda \dots \text{dB vlnová délka} = \frac{h}{\sqrt{2\pi m k_B T}}$$

$$\begin{aligned} \frac{P}{k_B T} &= \Lambda^{-3} g_{5/2}^\pm(z) \approx \Lambda^{-3} \left[z \pm \frac{z^2}{2^{5/2}} + \dots \right] \\ P &= \Lambda^{-3} g_{3/2}^\pm(z) \approx \Lambda^{-3} \left[z \pm \frac{z^2}{2^{3/2}} + \dots \right] \end{aligned} \quad \leftarrow \text{vyjádříme pomocí } g$$

$$z = g \Lambda^3 + a (g \Lambda^3)^2 \Rightarrow 0 = a \pm \frac{1}{2^{3/2}} = a = \mp \frac{1}{2^{3/2}}$$

$$\frac{P}{k_B T} = \Lambda^{-3} \left[g \Lambda^3 \pm \left(\frac{1}{2^{5/2}} - \frac{1}{2^{3/2}} \right) g^2 \Lambda^6 \right]$$

$$\frac{P}{k_B T} = g \left[1 \mp \frac{1}{2^{5/2}} g \Lambda^3 \right]$$

• klasická limita: $g \Lambda^3 \ll 1$

• korekce $\begin{cases} \text{kladné pro fermiony} \\ \text{záporné pro bosony} \end{cases}$